IT4010 RESEARCH PROJECT • PP1

OVERALL PROJECT:

AI & IoT-Based Smart Restaurant Management System
for Food Waste Prediction and Operational Optimization

MY INDIVIDUAL COMPONENT (Component 3 of 4):

AI-Based Multi-Modal Food Spoilage Prediction and Quality Assessment

Student: [Your IT Number] | [Your Name]
Project ID: J26-IT-333 | Specialization: [Your Specialization]
Supervisor: [Supervisor Name]

Research Objective:

Develop a predictive food-spoilage detection system using IoT sensor data and computer vision with machine learning to estimate remaining shelf-life.

SLIIT
Sri Lanka Institute of Information Technology

- Computer Vision
- IoT Sensors
- Machine Learning
- Real-time Dashboard
- Food Safety

AI-Based Multi-Modal Food Spoilage Prediction and Quality Assessment

Specialization: [Your Specialization Field]

[IT Number] | [Student Name] | J26-IT-333

Practice thoroughly to ensure that each individual component is delivered within exactly 2 minutes.

AI-Based Multi-Modal Food Spoilage Prediction & Quality Assessment

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Individual Component Overview • Component 3 of 4

Research Problem & Impact

Restaurants globally discard 30–40% of food inventory due to inadequate spoilage detection.

Key pain points:
Manual inspection is subjective & inconsistent
No real-time multi-sensor integration
No food-category-specific shelf-life estimation
Single-modality systems miss critical spoilage cues

Impact: Revenue loss, food safety risks, and increased carbon footprint from food waste.

Limitations of Existing Systems

Single-modality
Vision-only or sensor-only systems — not fused

Generic models
No category-specific (fruit / vegetable / meat) profiles

Reactive, not predictive
Alert only after spoilage is visible — shelf-life not estimated

No edge-deployable CPU-only solution for restaurants

Proposed Solution

A multi-modal AI system that fuses:

Computer Vision pipeline
42-dim feature vector: color moments, biological pigment indices (NDVI, anthocyanin), texture, Hu moments, geometry

IoT Sensor Integration (ESP32)
Temperature, Humidity, VOC, CO_2 — real-time chamber monitoring

ML Ensemble (Random Forest / GBM)
Species classifier, spoilage classifier, mould detector, ripeness regressor — <100 ms CPU inference

Output: Spoilage probability, risk level, remaining shelf-life (RSL) with 90% CI, mould flag

Research Gap & Novelty

No existing system combines:

Multi-modal fusion (60% CV + 40% IoT)
Food-category-specific assessment profiles for fruits, vegetables, meat, fish, dairy
Biological pigment-based degradation indices (chlorophyll loss, anthocyanin shift)
Real-time RSL estimation with confidence intervals
Edge-deployable ESP32-CAM + FastAPI stack

This component fills the gap by integrating CV + IoT fusion with category-specific produce encyclopaedia (47 species).

Main Objective

Develop a predictive food-spoilage detection system using IoT sensor data and computer vision with machine learning to estimate remaining shelf-life in restaurant storage environments.

Sub-Objectives

1. Build multi-modal feature extraction pipeline combining CV pigment analysis and IoT sensor telemetry
2. Train food-category-specific ML classifiers for spoilage probability and RSL prediction
3. Integrate ESP32-CAM capture into a React/FastAPI real-time dashboard with alert thresholds
4. Validate RSL estimation accuracy across 47 produce species using a produce encyclopaedia knowledge base

AI-Based Multi-Modal Food Spoilage Prediction & Quality Assessment

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Proposed Solution, Prototype & Technical Contribution • Component 3 of 4

System Architecture & Workflow

Input Capture
Food image
(upload / ESP32-CAM)
+ IoT sensors
(Temp, Humidity
VOC, CO_2)

→

Pre-processing
Resize 224×224
BGR→LAB/HSV
Feature extraction
42-dim vector

→

CV Engine
Species ID
Spoilage class.
Mould detect.
Ripeness regress.

→

Sensor Fusion
40% IoT weight
60% CV weight
Fused RSL
calculation

→

ML Prediction
RF / GBM ensemble
<100 ms CPU
Spoilage prob.
90% CI RSL

→

Output Dashboard
Risk level
RSL hours
Alert / action
Recommendation

System Inputs

Vision:
Food image (camera upload, ESP32-CAM stream, URL)
Sensors:
Temperature (°C), Humidity (% RH), CO_2 (ppm), VOC (ppm) via Firebase
Category:
Fruits / Vegetables • Storage duration (inferred from metadata)

System Outputs

Spoilage probability (%)
Risk level: Low / Medium / Critical
Remaining Shelf-Life (RSL) with 90% confidence interval
Mould detection flag + defect bounding boxes
Surface & colour diagnostics (pigment index values)
Actionable storage recommendations

User Requirements Addressed

UR-01: Real-time food inspection via camera or image upload
UR-02: Automatic species identification without manual input
UR-03: Shelf-life estimation with confidence range
UR-04: Risk alerts (Low / Medium / Critical) for staff
UR-05: IoT sensor integration with live dashboard
UR-06: Produce encyclopaedia (47 species, 4-stage spoilage profiles)
UR-07: CPU-only deployment — no GPU required

[PLACEHOLDER: Insert screenshots of working prototype here]
• Vision Inspector UI
• Live IoT dashboard
• Spoilage report output

[PLACEHOLDER: Insert genuine user feedback / evaluation results here]

Design Excellence & Novelty

1. Multi-Modal Fusion Architecture
60% optical CV + 40% IoT sensor weighting — weighted fusion for RSL not found in existing restaurant-grade tools

2. Biological Pigment Indices
7 degradation indices: NDVI, anthocyanin index, mould score, chlorophyll loss — go beyond simple RGB thresholds

3. Produce Encyclopaedia
47 species with 4-stage spoilage progression, pathogen risk, storage standards — category-specific knowledge base

4. Edge-Deployable CPU Stack
FastAPI + React + ESP32-CAM + Firebase — no GPU, no cloud AI API cost — restaurant-viable hardware

5. RSL with Confidence Interval
Shelf-life hours output with 90% CI — actionable, not just a binary 'fresh/spoiled' flag

AI & IoT-Based Smart Restaurant Management System

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Commercialization & Sustainability • Overall Project (J26-IT-333 — All 4 Components)

Component 1 Menu Optimisation & Demand Forecasting	Component 2 Smart Ordering & Supplier Management	Component 3 ← THIS Food Spoilage Prediction & Quality	Component 4 Customer Behaviour & Waste Analytics

Customer Persona

Primary:
Restaurant owners & kitchen managers (mid-to-large scale, 50–500 covers/day)

Pain points:
High food-cost wastage (~15–20% of revenue), manual spoilage checks, no predictive alerts

Secondary:
Hotel chains, hospital kitchens, airline catering, cloud kitchens

Goal:
Reduce ingredient waste, pass hygiene audits, improve COGS

Market Size

Global Food Waste Management Market
[PLACEHOLDER: Insert verified market size figure from a cited source]

Restaurant Technology Market
[PLACEHOLDER: Insert verified restaurant tech market figure]

Target Segment — Sri Lanka:
[PLACEHOLDER: Insert local restaurant/hospitality industry data]

Note: Market figures above are placeholders. Cite authoritative sources (Statista, WHO, USDA) before presenting.

Sustainability

Environmental:
Reduces landfill food waste → lower GHG emissions from decomposing organics
ESP32-based IoT — ultra-low-power edge hardware

Economic:
Reduces ingredient procurement cost for restaurants
CPU-only ML — no expensive GPU infrastructure

Social:
Improved food safety → reduced consumer illness risk
Potential surplus redistribution partnerships (NGOs, food banks)

What Makes This Unique?

End-to-end integration of 4 AI sub-systems on a single platform
Multi-modal food spoilage: CV + IoT fusion (not available in competitors)
Produce encyclopaedia with 47 species, 4-stage spoilage profiles
Edge-deployable — no GPU, no expensive API subscriptions
Category-specific shelf-life estimation with confidence interval
Restaurant-tailored — not a generic food-safety application

Commercialization Strategy

SaaS / Subscription Model
Monthly/annual licence per outlet — dashboard + API access + analytics

Hardware Bundle
ESP32-CAM sensor kit + cloud onboarding service

Enterprise / API
White-label API for POS / ERP vendors (hotel chains, cloud kitchens)

[PLACEHOLDER: Insert pricing tiers when finalised]

Cost Recovery

Direct Revenue:
Subscription fees, hardware kit sales, enterprise API licences

Indirect Value:
Restaurants recover R&D investment through reduced food waste (estimated 10–25% waste reduction)

Research Grants / SLIIT Commercialisation:
[PLACEHOLDER: Insert applicable grants / IP commercialisation path]

Break-even: [PLACEHOLDER — do not estimate without data]